EXPERIMENT NO: 2

Date :

AIM : To use basic Unix commands such as ls , mkdir, rmdir, cd, cat, banner, touch, file, wc, sort, cut and grep.

PROBLEM STATEMENT: EXECUTE THE FOLLOWING BASIC UNIX COMMANDS WITH THEIR DIFFERENT

OPTIONS.

THEORY :

UNIX/Linux is a multiuser and multitasking operating system. It provides a Command Line Interface (CLI) through which users can interact with the operating system. Linux commands are used to create and manage files and directories, display information, search data, and perform text processing efficiently.

Requirements

* Ubuntu/ Red Hat Linux
* Telnet or Terminal
* User Login ID

Shorthand at the Command Prompt

/ - root directory

./ - current directory

./command_name - run a command in the current directory when the current directory is not on the path

../ - parent directory

~ - home directory

$ - typical prompt when logged in as ordinary user

- typical prompt when logged in as root or superuser

! - repeat specified command

!! - repeat previous command

^^- repeat previous command with substitution

& - run a program in background mode

Typical Dot Files

.bash_logout- file executed by bash shell on logout

.login - initialization file when user logs in

.logout - commands run when user logs out

Useful Files

/boot/vmlinuz- the typical location and name of the Linux kernel. In the Slackwaredistribution, the kernel is located at/vmlinuz

/dev/fd1 - second floppy disk drive/etc/HOSTNAME - stores the name of the host computer/etc/hosts - contains a list of host names and absolute IP addresses.

/etc/hosts.allow- hosts allowed (by thetcpd daemon) to access Internet services

/etc/hosts.deny - hosts forbidden(by thetcpd daemon) to access Internet services

Important Directories

/bin - essential UNIX commands such asls, etc. Should contain all binaries needed to boot the system or run it in single-user mode

/boot - files used during booting and possibly the kernel itself are stored here

/home - home directories of individual users

/lib - standard shared library files

/root - home directory for root

Commands

Commands	Description
1. cal 2007	- show the timetable of 2007
1. cat /proc/ cpuinfo	- show information CPU info

1. date	-	show system date
1. uname-r	-	show used kernel version
1. init0	-	shutdown system
1. logout	-	leaving session
1. cd /home	-	enter to directory '/ home'
1. cd ..	-	go back one level
1. cd ../..	-	go back two levels
1. cp file1 file2	-	copying a file
1. cp dir/* .	-	copy all files of a directory within the current work directory
1. ln file1 lnk1	-	create a physical link to file or directory
1. ls	-	view files of directory
1. ls -l	-	show details of files and directory
1. lstree	-	show files and directories in a tree starting from root
1. mkdirdir1	-	create a directory called 'dir1'
1. mkdirdir1 dir2	-	create two directories simultaneously
1. mkdir-p /tmp/ dir1/dir2	-	create a directory tree
1. pwd	-	show the path of work directory
1. rm-f file1	-	delete file called 'file1'
1. rmdirdir1	-	delete directory called 'dir1'
1. tree	-	show files and directories in a tree starting from root(1)
1. find / -name file1	-	search file and directory into root file system from '/'
1. locate *. ps	-	find files with the '.ps' extension - first run 'updatedb' command
1. whereishalt	-	show location of a binary file, source or man
1. which halt	-	show full path to a binary / executable
1. ls -lSr |more	-	show size of the files and directories ordered by size
1. passwd	-	change password
1. useradduser1	-	create a new user

1. userdel-r user1	- delete a user ('-r' eliminates home directory)
1. gzip file1	- compress a file called 'file1'
1. unzip file1.zip	- decompress a zip archive
1. zip file1.zip file1	- create an archive compressed in zip
1. cat file1	- view the contents of a file starting from the first row
1. head -2 file1	- view first two lines of a file

CONCLUSION : Students are advised to write conclusion on separate sheet.

REVIEW QUESTIONS:

1. Difference between /home and /root.
2. Write a command for creating two directories named as A1 and A2 simultaneously.
3. What is the difference between cd .. and cd ../.. ?
4. What is the use of ./command_name command.
5. Which command can be used to rename a file in Linux.?
6. Which command is used to create and display file in Linux?
7. What is the core of Linux Operating System?
8. What is the basic difference between UNIX and Linux Operating System?
9. Linux is which kind of Operating System?
10. Which Command is used to check the kernel Version ?